

3.2.5. Numpy: Copying and Viewing Arrays

01:43

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

```
Original array after modifying view: <original_array>  
View array: <view_array>
```

- After modifying the copy:

```
Original array after modifying copy: <original_array>  
Copy array: <copy_array>
```

Code:

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array
```

```
original_array = np.array(inputlist)
```

```
# Create a view
```

```
view_array = original_array.view()
```

```
# Create a copy
```

```
copy_array = original_array.copy()
```

```
# Modify the view
```

```
view_array[0] = 99
```

```
print("Original array after modifying view:", original_array)
```

```
print("View array:", view_array)
```

```
# Modify the copy
```

```
copy_array[1] = 88
```

```
print("Original array after modifying copy:", original_array)
```

```
print("Copy array:", copy_array)
```

Average time

0.023 s

23.00 ms

Maximum time

0.029 s

29.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

29 ms

Debug



Expected output

10 20 30 40 50 60 70 80

Actual output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

✓ Test case 2

26 ms

Debug



Expected output

99 2 3 4 5

Actual output

99 2 3 4 5

Original array after modifying view: [99 2 3 4 5]

Original array after modifying view: [99 2 3 4 5]

View array: [99 2 3 4 5]

View array: [99 2 3 4 5]

Original array after modifying copy: [99 2 3 4 5]

Original array after modifying copy: [99 2 3 4 5]

Copy array: [99 88 3 4 5]

Copy array: [99 88 3 4 5]